Total Points: [16] [U2HW19 - W29 - Naming Compounds 1]

Q1	A B C D E **F**
Q2	A B C D E **F**
Q3	A B C **D** E F
Q4	A B C D **E** F
Q5	A B **C** D E F
Q6	A **B** C D E F
Q7	A B C D E **F**
Q8	A B **C** D E F
Q9	A B **C** D E F
Q10	**A** B C D E F
Q11	A **B** C D E F
Q12	A B C **D** E F
Q13	A B **C** D E F
Q14	**A** B C D E F
Q15	**A** B C D E F
Q16	A B C D **E** F

1. If an acid is formed from a polyatomic ion named "carbonate" (CO_3^{2-}), what will be the name of the corresponding acid (H_2CO_3)? [1]

Hydrogen carbon trioxide acid

Hydrocarbonous acid

Percarbonic acid

Hydrocarbonic acid

Carbonous acid

Carbonic acid

2. Many polyatomic ions, such as chlorate (ClO_3^-) and phosphate (PO_4^{3-}), primarily consist of which types of elements bonded together before acquiring an overall charge? [1]

A metalloid and a noble gas

A metal and a nonmetal element

Only hydrogen and a metal

Two different metal elements

A transition metal and a halogen

Multiple nonmetal elements

3. Given that the sulfide ion is S^{2-}, what is the correct chemical name for the compound Fe_2S_3 (Iron can have multiple charges)? [1]

Ferrous(II) sulfite

Iron(II) trisulfide

Iron sulfide

Iron(III) sulfide

Diiron trisulfide

Iron(II) sulfide

4. The acid $HClO_2$ contains the chlorite ion (ClO_2^-). According to the rules for naming polyatomic acids (oxyacids), what is the correct chemical name for $HClO_2$? [1]

Perchloric acid

Hypochloric acid

Hydrochlorous acid

Hydrochloric acid

Chlorous acid

Chloric acid

5. A compound is named 'Lead (IV) oxide'. What does the '(IV)' indicate about the lead in this specific compound? [1]

Lead is in the fourth period of the periodic table.

There are four oxygen atoms bonded to lead.

The lead ion has a +4 charge.

The oxide ion has a -4 charge.

It is the fourth most common isotope of lead.

There are four lead atoms in the formula.

6. An atom or group of atoms that has gained electrons, resulting in a negative charge, is known as a(n) _____, and is typically formed from a _____ element: [1]

polyatomic ion, metalloid

anion, nonmetal

cation, metal

cation, nonmetal

isotope, noble gas

anion, metal

7. What is the correct chemical name for the simple ionic compound K_2O, formed from potassium and oxygen? [1]

Potassium(I) oxide

Potassic oxide

Dipotassium monoxide

Potassium peroxide

Potassium(II) oxide

Potassium oxide

8. A compound with the formula PCl_3 consists of phosphorus and chlorine (both nonmetals). Based on its composition, how would this compound primarily be classified and named? [1]

As an oxyacid, because it contains a nonmetal and chlorine.

As a simple binary ionic compound, with chlorine named "chlorate".

As a covalent compound, using numerical prefixes.

As a polyatomic ion, requiring a Roman numeral for phosphorus.

As a binary acid, using the "hydro-" prefix.

As an ionic compound, naming the cation and then the anion ending in "-ide".

9. In the systematic naming of binary covalent compounds like SO_2, which rule applies to the use of Greek prefixes? [1]

Prefixes are only used if both elements have more than one atom.

Roman numerals are used instead of prefixes for the first element.

'Mono-' is omitted for the first element, but always used for the second if there is only one atom.

'Mono-' is always used for the first element if there is only one atom.

The prefix 'uni-' is used instead of 'mono-' for the first element.

Prefixes are only used for the second element, never the first.

10. When naming an ionic compound composed of a metal and a polyatomic ion, such as Na_3PO_4 (where PO_4^{3-} is phosphate), how is the polyatomic ion part named? [1]

Its name is used as is, without alteration.

A 'hydro-' prefix is added to its name.

Its ending is changed to -ic if it originally ended in -ate.

Its ending is changed to -ide.

Its ending is changed to -ous if it originally ended in -ite.

Numerical prefixes (di-, tri-) are added based on its implied count in the formula.

11. What is the correct chemical name for the compound $Al(NO_3)_3$ (NO_3^- is the nitrate ion).? [1]

Aluminum nitrite

Aluminum nitrate

Aluminum(III) nitrate

Aluminic trioxide nitride

Nitrogen aluminate

Aluminum trinitrate

12. The systematic name for a binary acid, such as H_2Se (hydroselenic acid), always includes which of the following combinations of prefix and suffix for the nonmetal part? [1]

No prefix, -ous acid

hypo- prefix, -ite acid

No prefix, -ate acid

hydro- prefix, -ic acid

per- prefix, -ic acid

hydro- prefix, -ous acid

13. Consider the acid H_3PO_4 (containing the phosphate ion, PO_4^{3-}) and the acid HI. Which statement accurately describes a key difference in their naming conventions? [1]

Both are named by simply stating "hydrogen" followed by the anion name (e.g., hydrogen phosphate).

Both are named using the "hydro-" prefix because they start with hydrogen.

H_3PO_4 (from phosphate) is named phosphoric acid (no "hydro-"), while HI is named hydroiodic acid.

H_3PO_4 is named using the "hydro-" prefix, while HI is not.

H_3PO_4 is named perphosphoric acid, and HI is named hypoiodous acid.

H_3PO_4 is named phosphorous acid, and HI is named iodic acid.

14. In a simple ionic compound formed between a metal like Barium (Ba) and a nonmetal like Fluorine (F), what is the correct ending for the anion's name (fluorine part)? [1]

-ide

-ate

-ous

-ic

-ite

-ine

15. A key characteristic used to initially identify a compound as a potential acid from its chemical formula is the presence of: [1]

Hydrogen as the first element.

A metal as the first element.

Oxygen as the last element.

Carbon as the central atom.

A halogen as the first element.

A polyatomic ion as the only component.

16. What is the correct chemical name for the binary covalent compound N_2O_4? [1]

Nitrogen(IV) oxide

Dinittric tetroxide

Dinitrogen oxide

Nitrogen oxide

Dinitrogen tetroxide

Nitrogen tetraoxygen